

Haoran Wang

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Research Interests

My research focuses on building trustworthy foundation models, with an emphasis on improving factuality, safety, and privacy through test-time control. I develop inference-time methods for large language models and multimodal systems that enhance robustness, interpretability, and reliability without retraining. I am broadly interested in interdisciplinary collaborations that translate trustworthy AI research into high-impact applications such as automated fact-checking, agentic systems, and AI for social good.

Education

- Present **PhD, Computer Science**, *Emory University*, Atlanta, GA, USA.
Dissertation: Test-Time Approaches for Trustworthy Large Language Models
Advisor: [Dr. Kai Shu](#), Committee Member: [Dr. Li Xiong](#), [Dr. Carl Yang](#), [Dr. Xiangliang Zhang](#)
- 2021 **MS, Computer Science**, *University of Oregon*, Eugene, Oregon, USA.
Advisor: [Dr. Thien Huu Nguyen](#)
- 2019 **BS, Computer Science**, *Purdue University*, West Lafayette, IN, USA.

Book Chapters and Journal Papers

- [J1] [Haoran Wang](#), Xiong Xiao Xu, Philip S. Yu, Kai Shu. [Beyond Tokens: A Survey on Decoding Methods for Large Language Models and Large Vision-Language Models](#). In *Proceedings of ACM SIGKDD Explorations Newsletter 28.1* (2026).
- [B2] [Haoran Wang](#), Baixiang Huang, Kai Shu. [Automated Fact-Checking](#). Chapter in *Oxford Handbook of Misinformation and Disinformation*, Oxford University Press.
- [B1] Baixiang Huang, [Haoran Wang](#), Kai Shu. [Factuality of Large Language Models: An Adversarial Perspective](#). Chapter in *Online Trust and Safety: Tools to Combat Online Harms, Misinformation, and Malicious Content*, Taylor and Francis CRC Press.

Conferences

- [C12] [Haoran Wang](#), Xiong Xiao Xu, Baixiang Huang, Kai Shu. [Privacy-Aware Decoding: Mitigating Privacy Leakage of Large Language Models in Retrieval-Augmented Generation](#). *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2026)*.
- [C11] Yue Huang, Chujie Gao, Siyuan Wu, [Haoran Wang](#), Xiangqi Wang, Yujun Zhou, Yanbo Wang, Jiayi Ye, Jiawen Shi, Qihui Zhang, Yuan Li, Han Bao, et al. [On the Trustworthiness of Generative Foundation Models: Guideline, Assessment, and Perspective](#). In *Proceedings of The Fourteenth International Conference on Learning Representations (ICLR 2026)*.
- [C10] Xiong Xiao Xu, [Haoran Wang](#), Yueqing Liang, Philip S. Yu, Yue Zhao, Kai Shu. [Can Multimodal LLMs Perform Time Series Anomaly Detection?](#). In *Proceedings of the ACM Web Conference 2026 (WWW 2026)*.

- [C9] Baixiang Huang, Zhen Tan, **Haoran Wang**, Zijie Liu, Dawei Li, Ali Payani, Huan Liu, Tianlong Chen, Kai Shu. [Model Editing as a Double-Edged Sword: Steering Agent Ethical Behavior Toward Beneficence or Harm](#). *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI 2026)*.
- [C8] Yueqing Liang, Liangwei Yang, Chen Wang, Congying Xia, Rui Meng, Xiong Xiao Xu, **Haoran Wang**, Ali Payani, Kai Shu. [Benchmarking LLMs for Political Science: A United Nations Perspective](#). *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI 2026)*.
- [C7] **Haoran Wang**, Kai Shu. [Spatial-Aware Visual Program Guided Reasoning for Answering Complex Visual Questions](#). *Findings of the Association for Computational Linguistics: IJCNLP-AACL 2025*.
- [C6] Aman Rangapur, **Haoran Wang**, Ling Jian, Kai Shu. [Fin-Fact: A Benchmark Dataset for Multimodal Financial Fact Checking and Explanation Generation](#). *Companion Proceedings of the ACM Web Conference 2025 (WWW 2025)*.
- [C5] **Haoran Wang**, Aman Rangapur, Xiong Xiao Xu, Yueqing Liang, Haroon Gharwi, Carl Yang, Kai Shu. [Piecing It All Together: Verifying Multi-Hop Multimodal Claims](#). In *Proceedings of the 31st International Conference on Computational Linguistics (COLING 2025)*.
- [C4] **Haoran Wang**, Kai Shu. [Trojan Activation Attack: Red-Teaming Large Language Models using Steering Vectors for Safety-Alignment](#). In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management (CIKM 2024)*.
- [C3] Yue Huang*, Lichao Sun*, **Haoran Wang***, Siyuan Wu*, Qihui Zhang*, Chujie Gao*, et al. [TrustLLM: Trustworthiness in Large Language Models](#). In *Proceedings of the Forty-first International Conference on Machine Learning (ICML 2024)* (* indicates equal contribution).
- [C2] **Haoran Wang**, Kai Shu. [Explainable Claim Verification via Knowledge-Grounded Reasoning with Large Language Models](#). In *Findings of the Association for Computational Linguistics: EMNLP 2023*.
- [C1] **Haoran Wang**, Yingdong Dou, Canyu Chen, Lichao Sun, Philip S. Yu, Kai Shu. [Attacking Fake News Detectors via Manipulating News Social Engagement](#). In *Proceedings of the ACM Web Conference 2023 (WWW 2023)*.

Preprints

- [P6] Weizhi Zhang, and many other authors including **Haoran Wang**. [JARVIS-Bench: Benchmarking Personal Intelligence Agents on Long-Horizon Real-User Daily Traces](#). *Preprint 2026*.
- [P5] **Haoran Wang**, Li Xiong, Kai Shu. [Do LLMs Know What Is Private Internally? Probing and Steering Contextual Privacy Norms in Large Language Model Representations](#). *Preprint 2026*.
- [P4] Baixiang Huang, Limeng Cui, Jiapeng Liu, **Haoran Wang**, Zhuiyue Tan, Yutong Chen, Chen Luo, Yi Liu, Kai Shu. [Towards Effective Model Editing for LLM Personalization](#). *Preprint 2026*.
- [P3] **Haoran Wang**, Maryam Khalid, Qiong Wu, Jian Gao, Cheng Cao. [Confidence-Guided Fact-Checking with Large Language Models through Probabilistic Certainty and Consistency](#). *Preprint 2026*.
- [P2] Aman Rangapur, **Haoran Wang**, Kai Shu. [Investigating Online Financial Misinformation and Its Consequences: A Computational Perspective](#). *Preprint 2023*.
- [P1] Canyu Chen, **Haoran Wang**, Matthew Shapiro, Yunyu Xiao, Fei Wang, Kai Shu. [Combating Health Misinformation in Social Media: Characterization, Detection, Intervention, and Open Issues](#). *Preprint 2022*.

Research Experience

- May 2025 – **Applied Scientist Intern**, Amazon AGI, Bellevue, WA.
Aug 2025
 - Project: [Confidence-Guided LLM Fact-Checker](#)
 - Mentor: Maryam Khalid, Qiong Wu, Jian Gao
 - Manager: Cheng Cao
- Spring 2025 – **Graduate Research Assistant**, Emory University, Atlanta, GA.
Present
 - Advisor: Kai Shu
 - Project: [DHS-CAOE](#), sponsored by DHS.
 - Developed efficient methods to mitigate hallucination in LLMs and LVLMs.
- Fall 2022 – **Graduate Research Assistant**, Illinois Institute of Technology, Chicago, IL.
May 2024
 - Advisor: Kai Shu
 - Project: [GUISE](#), sponsored by Charles River Analytics, DARPA.
 - Developed systems to extract information flows on social media using a hierarchical template approach.
- Fall 2018 – **Undergraduate Research Assistant**, Purdue University, West Lafayette, IN.
Spring 2019
 - Advisor: Yung-Hsiang Lu
 - Project: [CAM2](#), sponsored by NSF.
 - Evaluated different solutions to Big Data storage problem of unstructured data.
 - Built a distributed database to store images and videos along with their metadata captured by network cameras around the globe.

Fellowships & Awards

- 2026 🏆 **Google Cloud Research Credits**, Google
- 2026 🏆 **Excellence in Research Assistance Award**, Emory University
- 2025 🏆 **Research Access Program Award**, OpenAI
- 2022 🏆 **Provost Doctoral Fellowship**, Stevens Institute of Technology

Open-source Software

TrustGen (contributor): A modular and extensible toolkit for comprehensive trust evaluation of generative foundation models, (100+ Github ★)

TrustLLM (contributor): Trustworthy LLM Benchmark and Toolkit, (500+ Github ★)

Fin-Fact (contributor): Multimodal Financial Fact-Checking Dataset, (Benchmark dataset for shared task at Financial Misinformation Detection workshop at COLING 2025)

Teaching Experience

CS 570: Data Mining, April 20, 2026, Emory University.

Guest Lecturer

CS 585: Natural Language Processing, October 26, 2023, Illinois Institute of Technology.

Guest Lecturer

CS 550: Advanced Operating Systems, Fall 2024, Illinois Institute of Technology.

Graduate Teaching Assistant

CS 211: Computer Science I, Spring 2020, Winter 2021, Spring 2021, University of Oregon.

Graduate Teaching Assistant

CS 212: Computer Science II, Fall 2020, University of Oregon.

Graduate Teaching Assistant

Grant Proposal Writing Experience

NSF: SaTC: Countering Disinformation Risks in the Era of Large Language Models.

My Role: Discussing research objectives and writing multiple proposal tasks in interventions for influence risks posed by disinformation generated by large language models.

DHS: CAO: *Countering Misinformation in the Era of Large Language Models.*

My Role: Discussing research objectives and writing multiple proposal tasks in explainable fact-checking using large language models.

ASFOR: YIP: *Influence Narratives Defender: Leveraging Social Media Research for Detection, Interpretation, and Assessment of Influence Narratives.*

My Role: Formulating the research topic and writing proposal task, *Narrative Representation and Extraction*, *Narrative Detection and Interpretation*, and *Impact Assessment and Mitigation*.

Academic Service

Area Chair: NeurIPS Position Track {2025, 2026}

Program Committee: NeurIPS {2025, 2026}, ICML {2025}, AAAI {2024, 2025, 2026}, KDD {2024, 2025, 2026}, ACL {2025,2026}, PAKDD{2025}, ACML {2025}, ICLR {2026}, COLM {2026}

Journal Reviewer: Big Data Research, Neural Networks (NEUNET)

External Reviewer: SIGIR {2023, 2024}, WSDM{2023}, ICDM{2023}, PAKDD{2024}, SDM{2025}

Session Chair: CIKM 2024

Student Volunteer: ACM FAccT 2023

Mentoring

Iris Qiao, Undergraduate student at Emory University → M.S., Carnegie Mellon University

Aman Rangapur, IIT MS student → Solutions Architect, Allen Institute for AI (AI2)

Technical Skills

Programming languages: Python, Java, C, C++, C#, JavaScript, SQL, Bash, R, Julia

Deep learning frameworks: PyTorch, Hugging Face Transformers, PyTorch Geometric

HPC: CUDA, OpenMP, MPI

Software: Linux, Git, Google Cloud Computing, $\text{\LaTeX}$